

Project 2 - Graph Database Design and Cypher Query



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This is an **individual** project. The project contributes **20%** to the total assessment of this unit. The submission deadline is **15 May at 11:59 PM**.

This project is designed to practise solving business problems using graph database modelling techniques on a real-world dataset.

Datasets and Problem Domain

Please refer to the "Datasets and Problem Domain" section of Project 1.

Data Source


8MB

Project2_dataset.
csv

 Download

 Open

The dataset was preprocessed by removing certain columns and combining multiple CSV files. The original data source is available [here ↗](#).

Graph Database Design, Implementation and Queries

1. Design a property graph for the provided dataset, using the [Arrows App ↗](#) to draw the nodes and relations. Your design should be able to answer **ALL** questions in **item 3** below.

⚠ Note: The Arrows App has not been stable recently, but you can use the [alternative link ↗](#) to access it. Please note that the alternative Arrows App doesn't support opening a file from Google Drive, and you cannot save files to Google Drive with this version

2. Implement the property graph in Neo4j: Perform ETL to transfer the provided dataset into CSVs and then import these CSVs into Neo4j.

3. Write Cypher queries to answer the following questions.

- a. List all distinct airline names where the airline's country is Australia.
- b. How many route records in the dataset are domestic, and how many are international? A route is domestic if the departure and arrival airports are in the same country/region, and international if they are in different countries/regions.
- c. In the given dataset, each row represents a service operated by a specific airline between two airports. Find the airport pair with the greatest number of such records. Treat $A \rightarrow B$ and $B \rightarrow A$ as the same airport pair. Use `WITH` in your query.
- d. Each route may be operated using one or more aircraft types (stored as a semicolon-separated list in the dataset). Find the top 5 airport pairs that are served with the greatest number of distinct aircraft types across all rows and all airlines. Treat $A \rightarrow B$ and $B \rightarrow A$ as the same airport pair, and count distinct aircraft types across all rows for that pair.
- e. There are no direct flights between Beijing (Beijing Capital International Airport) and Perth (Perth International Airport). Find all possible travel routes from Beijing to Perth where at most 3 hops (i.e., at most 3 ROUTE relationships) are traversed. Treat each unique sequence of airports visited as a distinct route. How many such distinct routes exist?

- f. **(CITS5504 students only)** Find the top 5 pairs of airlines that compete head-to-head on the greatest number of shared routes. Two airlines are considered competitors on a route if they both operate between the same two airports, regardless of direction (treat A→B and B→A as the same route). Return the airline pair names and the number of routes they share.
4. Write at least two meaningful Cypher queries that can be answered using this graph. One of your queries must use [APOC](#) ↗.

 **Performance Note:** The Neo4j engine might experience slow query performance with a large dataset, depending on your device's capabilities. If you experience long query time, consider reducing the dataset size by filtering out records, such as deleting records for specific State/Year, to enhance performance.

Cannot install Neo4j Desktop on your own device: Please use [Neo4j AuraDB](#) ↗ for this project.

Submission Requirement: Please submit both the original (unfiltered) CSVs and the filtered (smaller) CSVs in two separate folders. While you can run your Neo4j queries on the smaller CSVs for performance reasons, it's essential to provide the original CSVs for our reference and verification. You have the option to filter records either directly in the original CSVs, within Neo4j using Cypher commands like "DELETE", or through programming languages. **Please determine the most suitable option for your needs.**

In summary, the process:

1. Design your property graph;
2. Perform ETL to transform the provided dataset into CSV files using Excel or other programming language;
3. Import CSVs into Neo4j using Cypher;
4. Write and execute queries. If queries are very slow on your device, filter out some records and try all queries again. Submit both the original (unfiltered) CSVs and the filtered (smaller) CSVs in two separate folders. If your queries run efficiently without filtering, then there's no need to filter out records. In such cases, submit only the original (unfiltered) CSVs.

Submission Guidelines



In-text citations and end-text references are required in this project report, following the [IEEE referencing style](#) ↗.

1. Script Files

- You may use any programming language to create CSV files for nodes and relationships; please submit your scripts in a suitable format.
 - *If you are going to use Excel to create CSV files for nodes and relationships, please ignore the script submission requirement.*

2. CSV Files: The CSV files for populating your database.

 All script files and CSV files need to be zipped into a .zip file.

3. A PDF contains:

- Describe/Discuss the design and implementation process with references to graph data modelling techniques to support your design choices.
- A screenshot of your design created in the [Arrows app ↗](#).

 Note: The Arrows App has not been stable recently, but you can use the [alternative link ↗](#) to access it. Please note that the alternative Arrows App doesn't support opening a file from Google Drive, and you cannot save files to Google Drive with this version

- The code, screenshots and explanations for the ETL process.
 - *Please include key screenshots and/or code in your report, rather than all your code.*
- *If you filter out some data due to the performance issue, specify which data was filtered and explain how you filtered them out (e.g. the smaller dataset for question f).*
- Queries (graph database implementation, the given queries and your own queries) and their corresponding results.

- Some of the queries may have thousands of results. Please do not include the actual results in the PDF - discuss interesting ones with the aid of screenshots.
- The results can be captured using screenshots.
- Discuss the practical applications of Graph Data Science, with specific reference to the graph algorithm(s) most suitable for at least one chosen application. **(Note: implementation in Neo4j is not required for this project. You don't need to use the given dataset or the graph database you designed for this task.)**
- Graph databases such as Neo4j maintain metadata about the structure of the graph, including node labels, relationship types, and property keys.
 - (a) Explain why metadata is an important consideration when designing a graph database schema, using references, with examples where appropriate.
 - (b) Analyse how a developer or data engineer can leverage metadata to write more effective and efficient Cypher queries, using references, with examples where appropriate. For example, through `apoc.meta.schema()` or `db.schema.visualization()`.
- Student name and student number

 Submit 2 files via LMS

- A single PDF file, file name: <student ID>_<last name>.pdf
- A single .zip file that contains all script files and CSV files, file name: <student ID>_<last name>.zip

 If you do not follow the submission instructions — such as missing files, submitting files in an incorrect format, not using a .zip file for your submission, or not following the required file naming convention — 10 marks will be deducted from your final score.

Marking Scheme

Scheme	Marks
Graph database design using Arrows app ↗ .	5
Discussions of the design choices with pros and cons identified	5
ETL: describes the complete process of transforming the provided dataset to CSVs (including any filtering methods and specific steps taken), including key code, explanations and important screenshots in your PDF file.	5
Graph database Implementation with Cypher code, including code and key screenshots of database stats (#nodes, #relations, labels, and relationships) in your PDF file.	5
Correctly answering the specified questions using Cypher.	5
Your self-designed queries with Cypher.	5
Discuss how graph data science can be applied to at least one useful application	5
Discussion of metadata in the graph database, explaining what it reveals about the schema (labels, relationship types, property keys) and how it supports Cypher query development.	5
Report quality (the PDF file).	5

Total**45**

You can interpret the scale of marks as:

5 - Exemplary (comprehensive solution demonstrating professional application of the knowledge taught in the class with initiative beyond just meeting the project requirement)

4 - Proficient (correct application of the taught concepts with clear understanding demonstrated)

3 - Satisfactory (managed to meet most of the project requirements)

2 - Developing (some skills are demonstrated but need revision)

1 - Not yet Satisfactory (minimal effort)

0 - Not attempted.

Can I Use Generation AI?

Informed and educated use of generative AI tools is encouraged and should be declared. If you choose to use generative AI, please document its suggestions in your report and provide a clear rationale for accepting or rejecting each recommendation. While using generative AI is optional, as a data science professional, you are strongly encouraged to develop critical evaluation skills and build familiarity with modern tools that could enhance your productivity.



NOTE: Using generative AI without proper citation and documentation is unacceptable.

Last updated 2 days ago